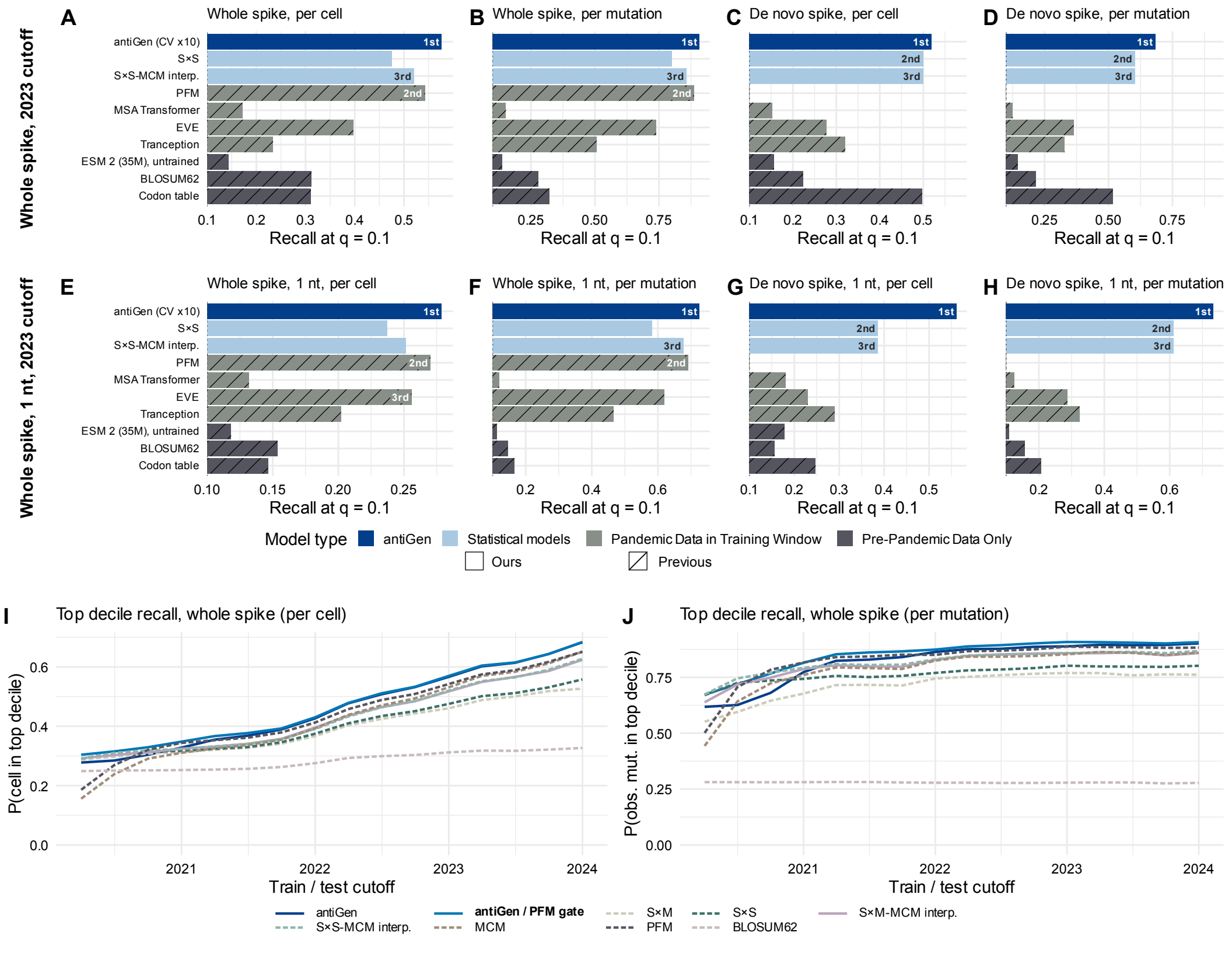
A
Whole spike, per cell
B
Whole spike, per mutation
C
De novo spike, per cell
D
De novo spike, per mutation
Whole spike, 2023 cutoff
antiGen (CV x10)
S×S
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
ESM 2 (35M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
Recall at q = 0.1
E
Whole spike, 1 nt, per cell
F
Whole spike, 1 nt, per mutation
G
De novo spike, 1 nt, per cell
H
De novo spike, 1 nt, per mutation
Whole spike, 1 nt, 2023 cutoff
Model type
antiGen
Statistical models
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
I
Top decile recall, whole spike (per cell)
P(cell in top decile)
Train / test cutoff
J
Top decile recall, whole spike (per mutation)
P(obs. mut. in top decile)
antiGen
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
MCM
PFM
BLOSUM62